

Will New Driving Technologies Change the Value of Public Transportation Investments?

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Road-map

- 1 Motivation
- 2 Data
- 3 Model
- 4 Counterfactuals
- 5 Implications

Motivation: should we revisit transit investments?

- Real capital spending on mass-transit ↑ **13-fold** since 1956.
- Costs/benefits recouped over decades.
- Self-driving vehicles (SDVs) ↓ opportunity cost of in-car time.
- Tension?: *Transit megaprojects vs. a future dominated by SDVs.*
- Boston area as example context.



Autonomous-driving technology

- SAE Levels 0–5: market now at 2–4.
- Mercedes Level 3 approved (CA, 2023); Waymo Level 4 robo-taxis (AZ, CA, TX).
- Follow Varian Rule?^a Mainstream in a decade.



SAE J3016™ LEVELS OF DRIVING AUTOMATION™

Learn more here: [sae.org/standards/content/j3016_202104](https://www.sae.org/standards/content/j3016_202104)

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SAE LEVEL 0*	SAE LEVEL 1*	SAE LEVEL 2*	SAE LEVEL 3*	SAE LEVEL 4*	SAE LEVEL 5*
You are driving whenever these driver support features are engaged – even if your feet are off the pedals and you are not steering.			You are not driving when these automated driving features are engaged – even if you are seated in “the driver’s seat”.		
You must constantly supervise these support features; you must steer, brake or accelerate as needed to maintain safety.			When the feature requests, you must drive.	These automated driving features will not require you to take over driving.	

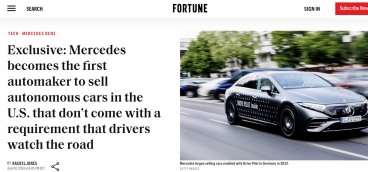
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These are driver support features			These are automated driving features		
These features are limited to providing warnings and momentary assistance.			These features can drive the vehicle under limited conditions and will not operate unless all required conditions are met.		
These features provide steering OR brake/acceleration support to the driver.			This feature can drive the vehicle under all conditions.		
• automatic emergency braking • blind spot warning • lane departure warning			• lane centering AND adaptive cruise control at the same time • lane centering AND adaptive cruise control at the same time • traffic jam chauffeur • local driverless taxi • pedals/steering/wheel may or may not be installed		
Example Features			• same as level 4, but feature can drive everywhere in all conditions		

What does the human in the driver's seat have to do?

What do these features do?

Example Features



'22

Waymo launches rider-only service to the public in SF

In early 2022, we began serving fully autonomous rides – with no human behind the wheel – to employees and the public via through our website and to trusted riders.



^aSee

<https://www.theguardian.com/commentisfree/2015/apr/26/facebook-isnt-charity-poor-pay-by-surrendering-their-data>

Breaking News: Waymo announces strategic partnership with Toyota explore personally owned SDVs (April 29, 2025)

Public-transit investment landscape

- Boston Area's (MBTA) Considered investments (\$55 bn):
 - \$25 bn to return to state-of-good-repair.
 - \$30 bn upgrades proposed: improving electrification & frequency upgrades.
- Equivalent to $\approx$ \$8,000 per MA resident.
- Question: will riders (and revenues) materialise if/when SDVs are widely adopted?

Alternative 6

What if you could catch an electrified Commuter Rail train at nearly any station every 15 minutes at any time of day?

Cost: \$28.9 billion

*\$40.7 billion adjusted for 2030 inflation

Station Type	Typical Frequency ¹
Inner Core Stations	Every 15 min
Key Stations	Every 15 min
All Other Stations	Every 15 min where possible



¹In both directions

RAIL VISION

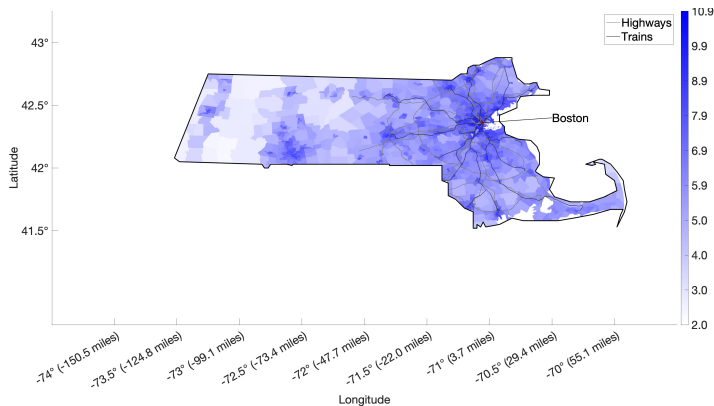
Massachusetts Bay Transportation Authority

- ① **Behavioural:** How do SDVs affect residential location, mode choice and vehicle miles?
- ② **Returns:** Do SDVs erode the marginal benefit of transit improvements?
- ③ **Policy:** Can faster / more convenient transit offset SDV impacts?

- **LEHD LODES** (2018): # workers by home/work census tract.
- **Travel times**: MBTA Trip Planner + OpenRouteService, 9 AM arrival (rush hour).
- **Prices**: MBTA fares; IRS variable driving cost (\$0.18/mi) + parking.
- **Sample**: 1,301 home tracts (≤ 75 mi), 100 work tracts (top 100 within 20 mi).

Residential Locations

Figure: Population Density by Census Tract



- Note, higher densities around highways and train lines

Summary statistics

	N	Mean	SD	Median	Min	Max
<i>Public transit only</i>						
Duration (minutes)	52,132	74.41	42.83	66.00	0.00	281.00
# segments	52,132	2.18	1.23	2.00	0.00	8.00
Price (\$)	52,132	8.74	9.14	3.40	0.00	60.40
<i>Driving</i>						
Duration	105,770	28.86	21.03	23.61	0.00	236.14
Price (\$)	105,770	25.20	15.32	25.84	0.00	83.35

- Driving is typically faster but more expensive.
 - Only 0.2% of tract pairs faster by transit.
 - 12% transit use suggests value in hands-free commute time.
- Drive-to-transit option omitted from table for brevity.

Modeling intuition

- Work location exogenous (supported by Barwick et al., 2024).
- Workers choose home h and commute mode $c \in \{\text{car, transit, both, outside good}\}$.
- Transport attributes impact location utility.
- Can infer preferences for commute characteristics from home location choices.
- Later simulate home and commute mode choices with SDVs.
 - Opportunity cost of time akin to hands-free public transit.

Utility specification

Commuting choice

$$U_{ichw} = X_{chw}\beta_g + \varepsilon_{ichw}$$

$$\lambda_{ghw} = \ln\left(1 + \sum_c e^{X_{chw}\beta_g}\right)$$

Home location

$$\mathbb{U}_{ighw} = \delta_{gh} + \lambda_{ghw} + \eta_{ihw}$$

$$s_{ghw} = \frac{\exp(\delta_{gh} + \lambda_{ghw})}{1 + \sum_{h' \in H} \exp(\delta_{gh'} + \lambda_{gh'w})}$$

- X_{chw} : active driving minutes, passive transit minutes, transfers, cost.
- Estimate β_g (low/high income) via ML.
 - Simultaneously recover δ_{gh} via contraction mapping (Berry et al., 1995).

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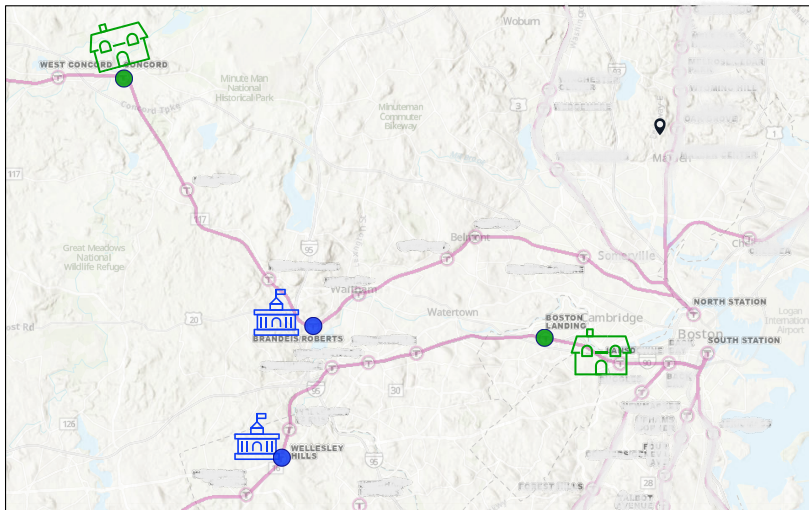
- **Mean utilities:**

- Estimate δ_{gh} via the Berry et al. (1995) contraction mapping
- δ_{gh} absorbs all location amenities (schools, crime, housing cost, etc.)

- **Transportation tastes.** The degree to which workers select residential locations with particular transportation characteristics identifies the strength of their preferences for those features.

- active travel time (car),
- hands-free travel time (transit),
- number of transfers,
- price.

Identification example (1)



Identification example (2)

- Similar drive times.
- By transit:
 - Concord faster to Brandeis,
 - Boston Landing faster to Wellesley.

<i>Transit travel times</i>		
Destination	Origin	
	Concord	Boston Landing
Brandeis	20 min	65 min (2 transfers)
Wellesley College	146 min (2 transfers)	30 min

- Brandeis workers disproportionately living in Concord implies preference for convenient transit

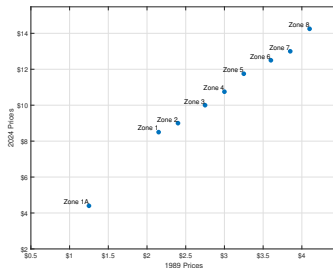
Price exogeneity in our setting

• Residential location

- Estimate location-specific utilities → absorb all observed & unobserved attributes.
- No need to isolate housing-price effects → classical price endogeneity does not arise.

• Transportation costs

- Private vehicles: gasoline (national market) independent of home choice.
- Public transit: MBTA fares are administratively set, not demand-optimized.
 - Bus/subway: flat fares.
 - Commuter rail: uniform zonal pricing; historical changes $\approx$ proportional across zones.



Commuter rail zone prices, 1989 vs 2024

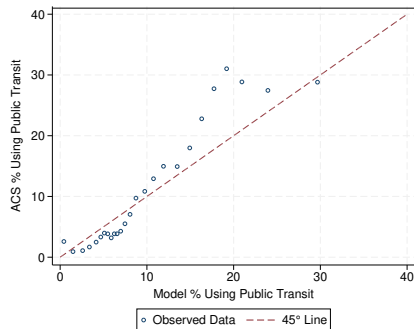
Table 2: Parameter estimates

	Households Earning:	
	$\leq \$3,333/\text{mo}$	$> \$3,333/\text{mo}$
Constant	5.084 (0.023)	5.385 (0.024)
1(Uses Transit)	-0.111 (0.028)	-0.470 (0.019)
Transfer Count	-0.210 (0.031)	-0.210 (0.020)
Driving Duration	-0.078 (0.000)	-0.070 (0.000)
Transit Duration	-0.051 (0.001)	-0.044 (0.001)
Price (\$)	-0.036 (0.002)	-0.018 (0.002)

- Active driving is more burdensome than hands-free travel:
 - **Low-income:** 1 driving minute $\cong$ 1.53 on-train minutes (0.078/0.051)
 - **High-income:** 1 driving minute $\cong$ 1.59 on-train (0.070/0.044)

Empirical Validation

- **Aggregate check.** Model predicts 12.2 % transit commuters vs. 2014-18 ACS figure of 11.1 % (same tracts).
- **Spatial pattern.** Binned-scatter (50 tracts per bin) of ACS share (y-axis) on model share (x-axis) shows close 45° alignment—see figure.
- **Station access mode.** Model: 61.4 % of commuter-rail riders drive/ride to the station. MBTA 2008-09 survey: 65.3 % arrive by private motorized modes (Humphrey et al., 2010).



Take-away: excellent external fit across three independent metrics, even though the estimation used no observed mode-choice data.

Changes/assumptions in counterfactuals

- SDVs: driving-time disutility = passive transit disutility; road capacity unchanged.
- Three housing-supply regimes: fixed locations, fixed stock, elastic supply.
- Transit improvements: onboard time reductions 20%, 40%, ...

Congestion modeling

- Zonal congestion: Underwood speed-flow $v = v_f e^{-k/k_0}$.
- Iterate speeds $\leftrightarrow$ location & mode choices until convergence.

Table 3: Impacts of self-driving vehicles

	Dynamic Congestion Effects			
	No SDVs	SDVs		
	Status Quo	Persistent Loc. Choices	Fixed Housing	Flexible Housing
E[Miles Driven (one way)]	9.53 (0.13)	11.80 (0.13)	12.47 (0.11)	13.13 (0.23)
E[Distance From Work]	14.38 (0.09)	14.38 (0.09)	15.12 (0.11)	16.51 (0.19)
E[Public Transit Rev/Worker]	0.80 (0.02)	0.69 (0.02)	0.70 (0.02)	0.73 (0.02)
E[Transit Rev/Rider]	6.53 (0.04)	6.14 (0.03)	6.30 (0.03)	6.45 (0.04)
% Take Transit	12.24 (0.27)	11.24 (0.24)	11.16 (0.23)	11.39 (0.24)

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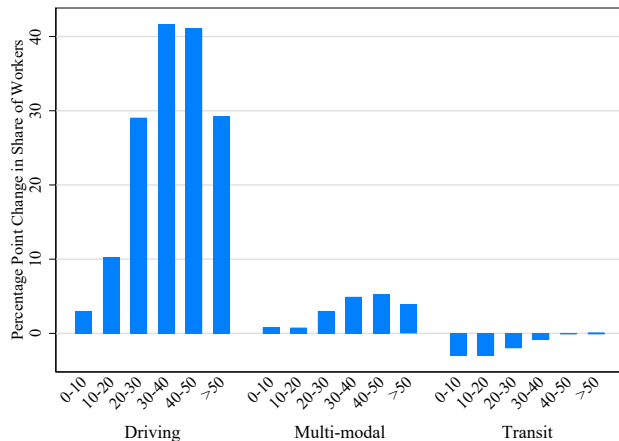
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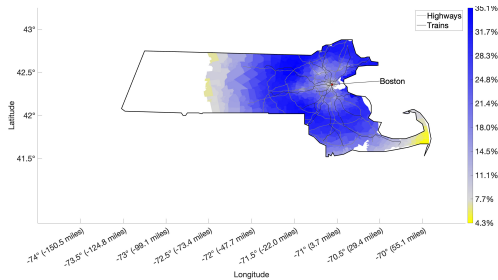
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Figure 4: Changes in transportation mode choices (persistent location)



Figures 5 & 6 combined

Price Changes–Inelastic Housing Supply



Population Changes–Elastic Housing Supply

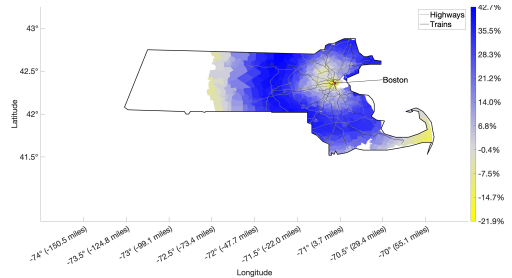


Table 4a: Impacts of public transit improvements on miles driven

	Miles Driven	
	No SDVs	SDVs
Status Quo Transit	0.00% (0.00)	37.85% (1.86)
20% Onboard Travel Time Reduction	-1.20% (0.07)	36.70% (1.85)
40% Onboard Travel Time Reduction	-2.83% (0.11)	34.98% (1.83)
60% Onboard Travel Time Reduction	-5.36% (0.18)	32.58% (1.88)
80% Onboard Travel Time Reduction	-8.72% (0.29)	29.26% (1.91)
100% Onboard Travel Time Reduction	-13.53% (0.44)	24.27% (1.98)

Table 4b: Impacts of public transit improvements on transit revenue

	Transit Revenue	
	No SDVs	SDVs
Status Quo Transit	0.00% (0.00)	-8.12% (0.54)
20% Onboard Travel Time Reduction	28.91% (0.32)	17.09% (0.87)
40% Onboard Travel Time Reduction	69.53% (0.91)	52.32% (1.50)
60% Onboard Travel Time Reduction	128.23% (1.95)	102.78% (2.65)
80% Onboard Travel Time Reduction	214.13% (3.86)	176.79% (4.63)
100% Onboard Travel Time Reduction	342.34% (7.26)	287.56% (8.11)

- Transit upgrades still attractive for revenue and usage.
- Weak lever for VMT/climate $\Rightarrow$ need congestion pricing, HOV, remote work.
- Government planning should incorporate SDV scenarios and housing-supply feedback.

Thank you for your comments!

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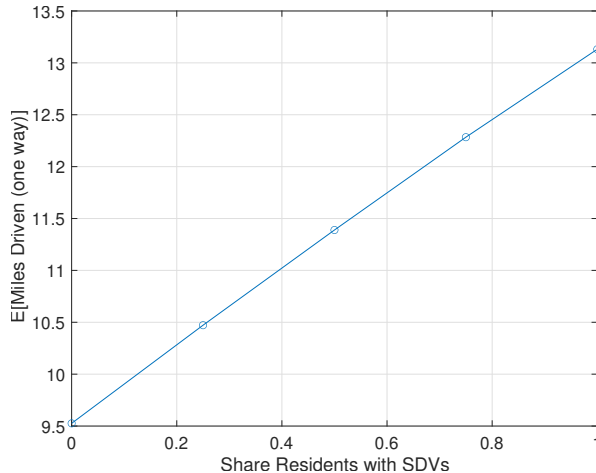
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Figure: Partial Self-Driving Penetration

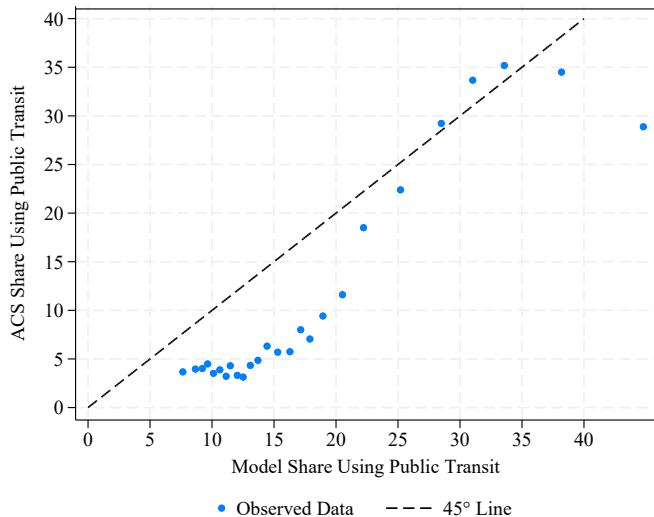


Alternative spec (no outside option): coefficient table

	Households Earning:	
	$\leq 3,333$ per month	$> \$3,333$ per month
1(Involves Public Transit)	0.325 (0.021)	-0.521 (0.018)
Transfer Count	-0.045 (0.014)	-0.083 (0.010)
Driving Duration	-0.060 (0.000)	-0.054 (0.000)
Transit Duration (Non-Auto)	-0.048 (0.001)	-0.043 (0.000)
Price (\$)	-0.027 (0.001)	-0.037 (0.001)

- Ratios of driving- vs. transit-time disutility fall to ~ 1.25 , but qualitative ordering unchanged.
- Fit to external data is poorer than baseline that does allow WFH etc. (next slide).

Fit check – no-outside-option model



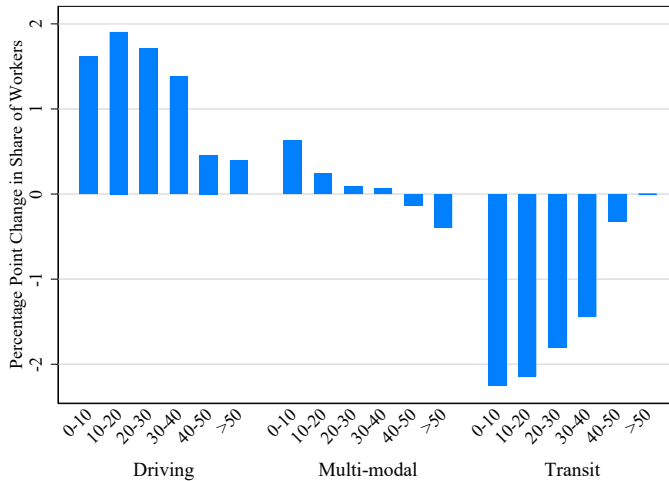
SDV results – no outside option

		Self-Driving Vehicles (SDVs)			
		Dynamic Congestion Effects			Congestion from Status Quo
		Persistent Location Choices	Fixed Housing Supply	Flexible Housing Supply	
Status Quo					
Transportation Outcomes					
E[Miles Driven (one way)]	15.39 (1.39)	15.60 (1.39)	16.18 (1.41)	19.43 (4.18)	22.91 (3.95)
E[Distance From Work]	16.31 (1.39)	16.31 (1.39)	16.91 (1.41)	20.18 (4.18)	23.55 (3.94)
E[Public Transit Revenue Per Worker/Day]	1.26 (0.02)	1.16 (0.02)	1.17 (0.02)	1.20 (0.02)	1.16 (0.02)
E[Public Transit Revenue Per Rider/Day]	6.14 (0.06)	6.02 (0.05)	6.06 (0.07)	6.08 (0.18)	5.99 (0.16)
Percent Take Public Transit	20.48 (0.30)	19.28 (0.24)	19.29 (0.24)	19.70 (0.35)	19.40 (0.36)

Transit speed-ups – no outside option

	Miles Driven		Transit Revenue	
	No SDVs	SDVs	No SDVs	SDVs
Status Quo Transit	0.00% (0.00)	26.24% (11.83)	0.00% (0.00)	-4.69% (0.96)
20% Onboard Travel Time Reduction	-1.93% (0.09)	23.99% (11.87)	24.08% (0.30)	17.88% (1.48)
40% Onboard Travel Time Reduction	-4.61% (0.15)	21.32% (11.85)	55.93% (0.85)	47.68% (2.28)
60% Onboard Travel Time Reduction	-7.86% (0.28)	17.57% (11.85)	98.58% (1.88)	87.73% (3.54)
80% Onboard Travel Time Reduction	-12.29% (0.40)	13.15% (11.80)	156.64% (3.70)	142.04% (5.72)
100% Onboard Travel Time Reduction	-17.72% (0.62)	7.15% (11.78)	236.05% (7.10)	216.59% (9.20)

Mode-choice shifts (no outside option)



Great-circle distance breakdown

Figure: Illustration of Great-Circle Paths Within Concentric Zones

Figure: *

Panel A: Home and Work Outside Circle

